Quantum Computing - Assignment 2

Simon's Algorithm: Implementation and Analysis

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Github Link - https://github.com/Sujoydey29/Quantum-Computing-Assignment-2/

Part A - Understanding the Reference Implementation

The reference implementation demonstrates Simon's Algorithm for the secret string `s = 011`. The algorithm uses n = 3 qubits for the query register and n = 3 qubits for the ancilla register, giving 6 qubits total, with 3 classical bits for measurement.

Role of Each Component

1. First Hadamard Layer

Applied to all n qubits in the query register at the start, the Hadamard gates transform each qubit from $|0\rangle$ into an equal superposition of $|0\rangle$ and $|1\rangle$. Collectively, this creates a uniform superposition of all 2^n possible n-bit input states simultaneously:

$$|0\rangle^{\otimes n} \rightarrow \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle$$

This allows the oracle to be evaluated on all inputs at once, which is the key quantum advantage - a classical computer would need to evaluate f(x) one input at a time.

2. The Oracle

Oracle encodes the secret string `s` by implementing the function $f(x) = f(x \oplus s)$ (2-to-1 function) mapping where each function value is shared by exactly two inputs.

The oracle is built in two stages:

- Copy step: CNOT gates copy the query register to the ancilla register - this creates $f(x) = x$ for all x.
- Collision step: For the index m of the first '1' in the reversed secret string, additional CNOT gates are applied from qubit m to all ancilla qubits corresponding to '1' positions in s. This XORs in the secret pattern and enforces $f(x) = f(x \oplus s)$.

For s = 011 (reversed: 110), m = 0. CNOT gates are applied from qubit 0 to ancilla qubits 3 and 4 - the positions of the two '1' bits in the reversed string.

3. Second Hadamard Layer

After the oracle, a second layer of Hadamard gates is applied to the query register. This performs quantum interference: states that do NOT satisfy $u \cdot s = 0 \pmod 2$ destructively cancel out, while states that DO satisfy this equation constructively reinforce.

The combined first-Hadamard → oracle → second-Hadamard operation transforms the state such that only the valid u vectors survive. Mathematically, the amplitude of any state |u⟩ in the output is proportional to:

$$\sum_x (-1)^{u \cdot x} \; |f(x)\rangle \;\rightarrow\; \text{non-zero only when } u \cdot s = 0 \pmod 2$$

4. Measurement

Only the query register (qubits 0 to n−1) is measured. Each shot produces a bit string u. Because of the interference created by the second Hadamard layer, only strings satisfying $u \cdot s = 0 \pmod 2$ are ever measured - all others have zero probability amplitude.

Repeating the circuit O(n) times yields enough independent equations to reconstruct s classically.

Why Measurement Outputs Satisfy $u \cdot s = 0 \pmod 2$

The mathematical reason follows directly from the structure of the quantum state after the second Hadamard layer. After the oracle entangles the query and ancilla registers, the full state is:

$$|\psi\rangle = \boxed{/}\{1\}\{\sqrt{\{2\text{^}n\}}\} \sum_x \, |x\rangle|f(x)\rangle$$

Since f satisfies the Simon promise $\big(f(x) = f(x \oplus s)\big)$, for each function value v, exactly two inputs $x \text{ and } x \oplus s$ map to it. After applying the second Hadamard to the query register and projecting onto a particular ancilla state $|v\rangle$, the amplitude of $|u\rangle$ in the query register is:

$$\propto (-1)^{u \cdot x} \, [1 + (-1)^{u \cdot s}]$$

This amplitude is zero unless $(-1)^{u \cdot s} = 1, \text{ i.e., } u \cdot s = 0 \pmod 2$. Therefore, only strings u orthogonal to s (mod 2) can ever be observed.

Verification for s = 011

For s = 011, the constraint u · s = 0 (mod 2) becomes: $\boldsymbol{u[1] \oplus u[0] = 0}$ (the two rightmost bits must be equal). This yields exactly 4 valid strings:

Bit String u	Valid? $(u[1] \oplus u[0] = 0)$
000	✓ Yes
011	✓ Yes
100	✓ Yes
111	✓ Yes

Simulation output: `{'011': 518, '000': 492, '111': 499, '100': 491}` - only these 4 strings appeared in 2000 shots, confirming the algorithm is correct.

Part B - Simon's Algorithm for s = 10110

The algorithm is now implemented for the 5-bit secret string `s = 10110`. With n = 5, the circuit requires 10 qubits (5 query + 5 ancilla) and 5 classical bits.

Oracle Construction for s = 10110

The oracle is built using the same scalable logic as Part A, accounting for Qiskit's little-endian qubit ordering:

- Copy step: $\mathrm{cx}(i,\ i+5) \quad \text{for } i = 0 \ldots 4$ - copies query register to ancilla.
- Collision $\mathrm{cx}(1,6),\ \mathrm{cx}(1,7),\ \mathrm{cx}(1,9)$ - for the three '1' positions in s_rev at indices 1, 2, and 4.

Circuit Execution

The 10-qubit circuit was executed using the Qiskit Aer simulator with 2000 shots. The expected number of valid measurement outcomes is $2^{(n-1)} = 2^4 = 16$ (including the trivial 00000).

Measured Bit Strings (excluding trivial 00000)

The following 15 non-trivial bit strings u were obtained from the simulation. All satisfy $u \cdot s = 0$ (mod 2):

u_1	u_2	u_3	u_4	u_5	u_6
00001	01000	01001	00110	00111	01110
01111	10010	10011	11010	11011	10100
10101	11100	11101			

All 16 expected valid strings (including 00000) appeared in the simulation, confirming the oracle is correctly implemented.

Part C - Classical Post-Processing (6 Marks)

Each measured bit string u satisfies the inner product equation $\boldsymbol{u \cdot s = 0}\ (\mathbf{mod}\ \mathbf{2})$. Collectively these form the matrix system $\boldsymbol{Us = 0}\ (\mathbf{mod}\ \mathbf{2})$, where U is the matrix, whose rows are the measured u strings.

System of Linear Equations

Writing $s = (s_4, s_3, s_2, s_1, s_0)$, each measured string gives one equation. The full set of 15 equations is:

u string	Equation (u · s = 0 mod 2)	Simplified
00001	$0 \cdot s_4 + 0 \cdot s_3 + 0 \cdot s_2 + 0 \cdot s_1 + 1 \cdot s_0 = 0 \quad s_0 = 0$	$0 \cdot s_4 + 0 \cdot s_3 + 0 \cdot s_2 + 0 \cdot s_1 + 1 \cdot s_0 = 0 \quad s_0 = 0$
01000	$0 \cdot s_4 + 1 \cdot s_3 + 0 \cdot s_2 + 0 \cdot s_1 + 0 \cdot s_0 = 0 \quad s_3 = 0$	$0 \cdot s_4 + 1 \cdot s_3 + 0 \cdot s_2 + 0 \cdot s_1 + 0 \cdot s_0 = 0 \quad s_3 = 0$
01001	$0 \cdot s_4 + 1 \cdot s_3 + 0 \cdot s_2 + 0 \cdot s_1 + 1 \cdot s_0 = 0 \quad s_3 + s_0 = 0$	$0 \cdot s_4 + 1 \cdot s_3 + 0 \cdot s_2 + 0 \cdot s_1 + 1 \cdot s_0 = 0 \quad s_3 + s_0 = 0$
00110	$0 \cdot s_4 + 0 \cdot s_3 + 1 \cdot s_2 + 1 \cdot s_1 + 0 \cdot s_0 = 0 \quad s_2 + s_1 = 0$	$0 \cdot s_4 + 0 \cdot s_3 + 1 \cdot s_2 + 1 \cdot s_1 + 0 \cdot s_0 = 0 \quad s_2 + s_1 = 0$
00111	$0 \cdot s_4 + 0 \cdot s_3 + 1 \cdot s_2 + 1 \cdot s_1 + 1 \cdot s_0 = 0 \quad s_2 + s_1 + s_0 = 0$	$0 \cdot s_4 + 0 \cdot s_3 + 1 \cdot s_2 + 1 \cdot s_1 + 1 \cdot s_0 = 0 \quad s_2 + s_1 + s_0 = 0$
01110	$0 \cdot s_4 + 1 \cdot s_3 + 1 \cdot s_2 + 1 \cdot s_1 + 0 \cdot s_0 = 0 \quad s_3 + s_2 + s_1 = 0$	$0 \cdot s_4 + 1 \cdot s_3 + 1 \cdot s_2 + 1 \cdot s_1 + 0 \cdot s_0 = 0 \quad s_3 + s_2 + s_1 = 0$
01111	$0 \cdot s_4 + 1 \cdot s_3 + 1 \cdot s_2 + 1 \cdot s_1 + 1 \cdot s_0 = 0 \quad s_3 + s_2 + s_1 + s_0 = 0$	$0 \cdot s_4 + 1 \cdot s_3 + 1 \cdot s_2 + 1 \cdot s_1 + 1 \cdot s_0 = 0 \quad s_3 + s_2 + s_1 + s_0 = 0$
10010	$1 \cdot s_4 + 0 \cdot s_3 + 0 \cdot s_2 + 1 \cdot s_1 + 0 \cdot s_0 = 0 \quad s_4 + s_1 = 0$	$1 \cdot s_4 + 0 \cdot s_3 + 0 \cdot s_2 + 1 \cdot s_1 + 0 \cdot s_0 = 0 \quad s_4 + s_1 = 0$
10011	$1 \cdot s_4 + 0 \cdot s_3 + 0 \cdot s_2 + 1 \cdot s_1 + 1 \cdot s_0 = 0 \quad s_4 + s_1 + s_0 = 0$	$1 \cdot s_4 + 0 \cdot s_3 + 0 \cdot s_2 + 1 \cdot s_1 + 1 \cdot s_0 = 0 \quad s_4 + s_1 + s_0 = 0$
11010	$1 \cdot s_4 + 1 \cdot s_3 + 0 \cdot s_2 + 1 \cdot s_1 + 0 \cdot s_0 = 0 \quad s_4 + s_3 + s_1 = 0$	$1 \cdot s_4 + 1 \cdot s_3 + 0 \cdot s_2 + 1 \cdot s_1 + 0 \cdot s_0 = 0 \quad s_4 + s_3 + s_1 = 0$
11011	$1 \cdot s_4 + 1 \cdot s_3 + 0 \cdot s_2 + 1 \cdot s_1 + 1 \cdot s_0 = 0 \quad s_4 + s_3 + s_1 + s_0 = 0$	$1 \cdot s_4 + 1 \cdot s_3 + 0 \cdot s_2 + 1 \cdot s_1 + 1 \cdot s_0 = 0 \quad s_4 + s_3 + s_1 + s_0 = 0$
10100	$1 \cdot s_4 + 0 \cdot s_3 + 1 \cdot s_2 + 0 \cdot s_1 + 0 \cdot s_0 = 0 \quad s_4 + s_2 = 0$	$1 \cdot s_4 + 0 \cdot s_3 + 1 \cdot s_2 + 0 \cdot s_1 + 0 \cdot s_0 = 0 \quad s_4 + s_2 = 0$
10101	$1 \cdot s_4 + 0 \cdot s_3 + 1 \cdot s_2 + 0 \cdot s_1 + 1 \cdot s_0 = 0 \quad s_4 + s_2 + s_0 = 0$	$1 \cdot s_4 + 0 \cdot s_3 + 1 \cdot s_2 + 0 \cdot s_1 + 1 \cdot s_0 = 0 \quad s_4 + s_2 + s_0 = 0$
11100	$1 \cdot s_4 + 1 \cdot s_3 + 1 \cdot s_2 + 0 \cdot s_1 + 0 \cdot s_0 = 0 \quad s_4 + s_3 + s_2 = 0$	$1 \cdot s_4 + 1 \cdot s_3 + 1 \cdot s_2 + 0 \cdot s_1 + 0 \cdot s_0 = 0 \quad s_4 + s_3 + s_2 = 0$
11101	$1 \cdot s_4 + 1 \cdot s_3 + 1 \cdot s_2 + 0 \cdot s_1 + 1 \cdot s_0 = 0 \quad s_4 + s_3 + s_2 + s_0 = 0$	$1 \cdot s_4 + 1 \cdot s_3 + 1 \cdot s_2 + 0 \cdot s_1 + 1 \cdot s_0 = 0 \quad s_4 + s_3 + s_2 + s_0 = 0$

Solving the System

A brute-force Python solver tests all 31 non-zero 5-bit candidates against the full set of equations. A candidate's s^* is accepted only if $u \cdot s^* = 0 \pmod 2$ holds for every measured u string:

```
n = 5
valid_u_strings = ['00001','01000','01001','00110','00111',
                   '01110','01111','10010','10011','11010',
                   '11011','10100','10101','11100','11101']

for i in range(1, 2**n):
    candidate = format(i, f'0{n}b')
    valid = all(
        sum(int(u[j])*int(candidate[j]) for j in range(n)) % 2 == 0
        for u in valid_u_strings
    )
    if valid:
        print('Recovered s:', candidate)
```

Result

Parameter	Value
Original secret string (s)	`10110`
Recovered secret string	`10110`
Match	✓ **`Perfect match`**
Equations used	`15 (all non-trivial u strings)`
Expected valid strings	$2^4 = 16$
Observed valid strings	`16 (including 00000)`

Conclusion

The classical post-processing successfully and uniquely recovered s = 10110 from the quantum measurement results. The uniqueness of the solution demonstrates that a sufficient number of linearly independent equations were collected - for an n-bit secret string, only n−1 = 4 linearly independent equations are needed, and the 15 equations obtained from 2000 shots easily satisfy this requirement.

This assignment demonstrates the full Simon's algorithm pipeline: quantum superposition → oracle evaluation → quantum interference → classical linear algebra, achieving exponential speedup over any classical approach.